



KERA-AWD Connect

Insights from research and field actions from **Kerala's** low-emission rice systems

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What gets measured gets managed – Peter Drucker

At the heart of the KERA-AWD initiative lies a simple but powerful idea—that careful measurement is the first step towards meaningful change. Across the project landscapes, efforts are underway to systematically capture not only greenhouse gas emissions such as methane and nitrous oxide, but also key indicators like crop yield, water use, and soil properties.

Together, these measurements create a more complete picture of the rice production system, linking environmental outcomes with farm performance. By grounding decisions in field-based evidence, the initiative is enabling a deeper understanding of how water and crop management practices influence both productivity and emissions. In doing so, KERA-AWD is helping pave the way for more informed, balanced, and climate-resilient agricultural practices.

Greetings on the occasion of Vishu. The new year presents an opportunity to build on our experiences, adopt necessary course corrections, and move forward with renewed focus towards achieving higher levels of efficiency, impact, and sustainability

and preferences regarding innovative low-emission rice cultivation practices that offer the dual benefits of reduced GHG emissions and enhanced agronomic outcomes.

Strengthening the evidence base

Capturing farmer insights, practices, and field experiences

a. From emissions to innovation: Detailed household surveys at pilot sites

With the primary objective of estimating greenhouse gas (GHG) emissions from rice using a Tier 3 methodology, a comprehensive household survey was conducted among 936 rice farmers across 17 pilot padasekharams. The data collected through this survey constitutes a critical component of the baseline assessment aimed at establishing a robust digital monitoring, reporting, and verification (MRV) system. In addition to quantifying emissions, the survey also aimed to capture farmers' awareness, exposure,



Fig 01. Detailed household survey training



Fig 02. Survey in the field

b. Integrating lean agricultural landscape survey into the KAU RAWE program

To strengthen evidence-based agricultural research and planning in Kerala, there is a growing need for regular and granular data on crop patterns and farm practices across diverse landscapes. The lean agricultural landscape survey is a structured data collection tool designed to capture farm-level information on climate-smart agricultural practices, with a particular focus on key crops in Kerala, such as rice, banana and coconut. With the objective of enhancing experiential learning and providing

practical exposure to data-driven agricultural development, the survey was integrated into the RAWE (Rural Agricultural Work Experience) programme of Kerala Agricultural University (KAU). The training programme was conducted across all four agricultural colleges of KAU, reaching a total of 454 students. Each trained student is expected to complete 15 household surveys, collectively contributing to an estimated dataset of approximately 6,810 farm-level observations across the state. The survey is currently in progress, with around 4,000 observations completed to date.



Fig 03. Training programme on lean agricultural landscape survey for the RAWE students

Enhancing Spatial Insights of the Landscape and Crops through Drones and Satellite Monitoring

Geospatial mapping and monitoring activities are progressing steadily to support precise assessment of padasekharams boundaries, generate the Digital Terrain Model (DTM), Digital Surface Model (DSM) datasets and paddy crop condition analysis. Drone mapping of experimental and control plots in Palakkad and Thrissur has been successfully completed, providing high-resolution spatial data(10cm). Climatic data extraction from IMD for Kerala has been completed. Second phase of drone mapping in Palakkad has been carried out, with data pre-processing of drone images in progress.



Fig 04. Drone mapping at Koorodumannam



Fig 05. Drone mapping at Thoppalam

Post-harvest drone surveys and mapping are planned for the Kole fields to capture end-stage field conditions. Ground truthing survey in Palakkad district for Land Use Land Cover (LULC) classification, paddy area validation and crop type delineation has been completed. It is currently progressing in Thrissur where these ground truth points will be used as training sites for calibration of satellite images and accuracy assessment of the geospatial outputs.



Fig 06. Ground truthing at Palakkad district

Systematic Evaluation Through Station-Based Trials

Station-based trials were continued at the Agricultural Research Station (ARS), Mannuthy, and the Regional Agricultural Research Station (RARS), Pattambi with systematic crop management and regular observations. Key operations included irrigation monitoring, fertilizer application and the implementation of plant protection measures. Regular observations on weed flora were conducted, along with soil and plant sampling. Biometric data were recorded at



Fig 07. Uprooting of plant for biometric

different crop growth stages– 30, 60, and 90 days after transplanting (DAT), as well as at the flowering stage–across various experiments, including amendment, varietal, and microbiome trials. Harvesting has been completed in two of the three experimental fields at ARS Mannuthy and observations on yield and

yield attributes have been recorded. Sample analysis is currently in progress. Harvesting and crop-cutting operations at RARS Pattambi are expected to be completed in April.



Fig 08. Gas sampling using Trace Gas Analyser (TGA)



Fig 09. Spraying clean rise on the crop



Fig 10. Crop cut at ARS, Mannuthy

From Field to Map: Digitizing Soil Properties

Geo-referenced soil samples collected from fields across the pilot sites are being analyzed for pH, electrical conductivity (EC), organic carbon, and both macro- and micronutrients. Standardized protocols and calibrated instruments are being employed to ensure accuracy and reliability. Analysis of pH and EC is currently



underway to assess soil reaction and salinity levels. The results will support the development of a comprehensive digital soil inventory, which will subsequently be integrated into landscape-level spatial modelling.

Fig 11. Soil sample analysis in progress

Measuring Emissions, Informing Action

A Gas Chromatograph for GHG emission analysis –the Shimadzu Nexis GC-2030 with HS-20–was officially installed in the last week of March and is expected to commence sample analysis from April onwards. This much-awaited instrument marks a significant milestone in the establishment of a dedicated GHG laboratory. At present, the instrument is housed at the Pesticide Residue Laboratory at KAU, Vellanikkara pending the establishment of the GHG laboratory.



Fig 12. Gas Chromatograph (Shimadzu Nexis GC-2030)

The system can analyze CO₂, CH₄, and N₂O in a single sample injection, enabling efficient and high-precision measurements. The Flame Ionization Detector (FID) is used for methane (CH₄) and carbon dioxide (CO₂) analysis, while the Electron Capture Detector (ECD) is employed for nitrous oxide (N₂O).

From Insights to Impact

On-Farm Trials in Action

On-farm trials were continued with regular monitoring and data collection activities in the Kole and Palakkad fields. Field conditions and crop growth stages were consistently monitored through regular field visits. In cases of suspected pest and disease attacks, appropriate management measures were advised. Weekly water measurements were carried out, encompassing both sensor-based and manual observations, as well as the monitoring of AWD pipes. Biometric data were recorded at different crop growth stages—30, 60, and 90 days after tra-

nsplanting (DAT), as well as at the flowering stage. Harvesting, along with one cent (0.01 acre) crop cuts, has been completed in both the Kole and Palakkad fields, and observations of yield and yield parameters have been recorded.

Fig 13. Taking water sensor reading in farmer's field



Fig 14. One cent crop cut in Palakkad field



Fig 15. Measuring the yield after crop cut at Thrissur



Fig 16. Measurement of grain moisture content

Improving Field-Level Decision Support Systems

Synthetic Aperture Radar (SAR) based Soil Moisture Mapping

A systematic soil moisture field data collection program has been initiated to support the calibration and validation of satellite-derived soil moisture products from the NASA-ISRO Synthetic Aperture Radar (NISAR) mission. A total of 22 sampling locations has been strategically selected in three padasekharams, representing both middle and tail-end of Malampuzha canal.

This ensured capturing the spatial variability in soil moisture conditions attributed to irrigation dynamics. Field measurements are being taken by employing state-of-the-art soil moisture sensors from Delta-T devices, with standardized protocols for depth, geolocation, and repeat observations. To ensure data quality and consistency, trainings were provided to field staff on the proper use of the instrument, data collection procedures, and regular time-series monitoring aligned with satellite overpass schedules. The collected in-situ data will serve as ground truth for calibrating and validating SAR-based soil moisture products, enabling the development of reliable, high-resolution soil moisture maps at regular intervals to support irrigation management and decision-making in rice-based systems.



Fig 17. Taking soil moisture measurements in the farmer fields

Smart Irrigation Insights from Field Sensors

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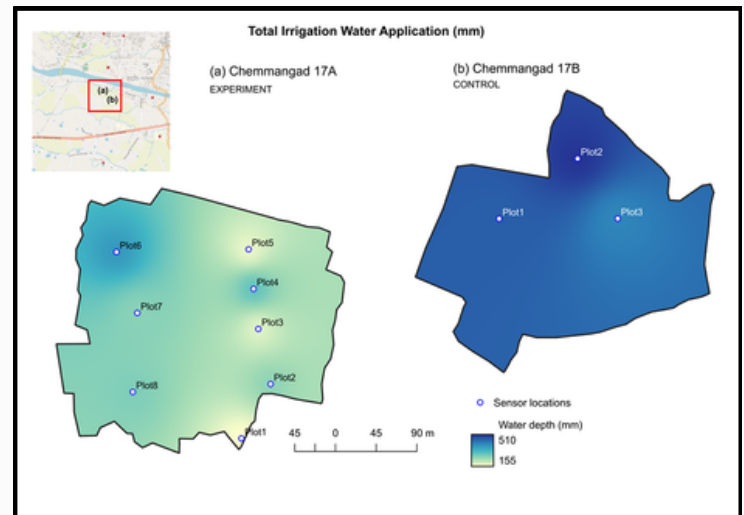


Fig 18. Total irrigation water application (in mm) across experimental (left) and control (right) padasekharams. Plots indicate the sensor locations within the padasekharams.

Building Capacity, Growing Futures

Bridging Landscapes: Lessons from Vietnam on Climate-Resilient and Market-Linked Agriculture

The KERA team's international exposure visit to Vietnam was an exceptionally enriching experience, offering valuable insights into innovative agricultural practices and emerging opportunities for collaboration. The visit provided a comprehensive understanding of Vietnam's dynamic agricultural ecosystem, particularly its strong emphasis on export-oriented cooperative models. Notably, cooperatives across scales demonstrated remarkable efficiency in managing diverse product portfolios while maintaining a clear focus on global markets.



Fig 19. KERA team in the Mekong Delta, Vietnam

A key highlight of the visit was the opportunity to closely engage with the Mekong Delta rice system, widely regarded as a global benchmark. The team gained deeper insights into the coordinated planning required for managing large and complex agro-ecosystems, alongside progressive initiatives such as the scaling of low-

carbon rice production, strategic branding efforts, and the evolving exploration of carbon credit mechanisms expected to mature over the coming years.

While further understanding is required—especially regarding the operationalization of MRV systems—the team identified strong parallels between the Mekong Delta and the Kolar lands under KERA. These similarities offer a promising foundation for contextual adaptation, knowledge exchange, and collaborative learning. The visit also facilitated meaningful interactions with key agribusiness stakeholders and entrepreneurs, particularly during the final day. These engagements opened avenues for potential partnerships with firms that are well-positioned to contribute to transformative developments in the agriculture sector.

Overall, the exposure visit significantly strengthened the team's perspective on sustainable, market-linked, and climate-resilient agricultural development, while laying the groundwork for future international cooperation.

Training on GHG Emissions Analysis

This activity focused on strengthening the technical capacity of KAU faculty and research staff to independently conduct GHG emission measurements in rice-based systems using Trace Gas Analyzer (TGA).



Fig 20. Training on GHG emissions analysis

The training combined structured lectures, hands-on instrument operation, field inspections, and technical discussions to ensure both conceptual understanding and practical competence. Lecture sessions covered the principles of GHG emissions in flooded rice systems, operational theory of portable gas analyzers, and detailed discussion of hardware and software TGA components, including analyzer configuration, instrument setup, data acquisition, file management, and quality assurance procedures. Hands-on training sessi-

ons allowed participants to directly operate CH₄ and N₂O analyzers under field conditions. Field inspections, dry runs, and a farmers' field visit were conducted to contextualize protocols within real experimental environments and to identify logistical constraints related to plot layout, accessibility, and analyzer mobility.

Training on Using Remote Sensing Techniques for Rice Growth and Soil Moisture

A one-day training programme was conducted featuring Dr. Narendra Das from Michigan State University, USA, Team Leader of the NASA NISAR Mission, with support from IRRI scientists. The training introduced the basics of radar remote sensing and its applications in detecting key indicators such as soil moisture, crop phenology, and planting dates, effectively engaging the participants and sparking their interest in the subject. Introductory sessions on Synthetic Aperture Radar (SAR) and the applications of NISAR L-band datasets were also covered, along with hands-on training.



Fig 21. Training on applications of remote sensing in rice crop

The participants included undergraduate (B.Sc. Agriculture) and postgraduate (M.Sc. and Ph.D.) students, as well as faculty members from the College of Agriculture, Vellanikkara; College of Forestry, Vellanikkara; and the College of Climate Change and Environmental Science, Vellanikkara, along with scientists from the Centre for Water Resources Development and Management (CWRDM).

Voices that matter



Mr Jacob Joy
(Technical Officer, State Project Management Unit, KERA)

“I am pleased to share our experience with the adoption of Alternate Wetting and Drying (AWD) in paddy cultivation across project areas. The introduction of AWD under KERA will significantly contribute to improving water use efficiency, which is crucial in the context of increasing water scarcity and climate variability. In Malampuzha region in Palakkad, where irrigation infrastructure will be strengthened through canal renovation, AWD will help farmers to optimize irrigation scheduling, reduce unnecessary water usage, and maintain crop productivity. Additionally, this practice has shown measurable benefits in reducing methane emissions from paddy fields, thereby contributing to climate change mitigation efforts.”



Mr Mejo George
(Joint Secretary, Onpathumri Padasekharam Samiti, Kolelands)

“The Kole wetlands are a Ramsar designated site, offering a several ecosystem services with paddy cultivation acting an ecological stabilizing role. However, our interaction with KAU & IRRI informed us that the current practice of continuous flooding is also a major contributor to global warming. We piloted the implementation of AWD and other soil amendments in our padasekharam this year. Our experience suggests that AWD can be implemented without yield loss in the Kole wetlands. Irrigation scheduling is a critical component of AWD implementation, and some financial incentives for repair and maintenance of field channels would accelerate its adoption”.



Ms Sruthi
(Agriculture Officer, Alathur Panchayat, Palakkad)

“By implementing the AWD (Alternate Wetting and Drying) practice under the KERA project, it has been possible to develop a more precise crop calendar for farmers, enabling timely and efficient rice cultivation. The adoption of practices such as biochar and phospho-gypsum application in the field has further helped in understanding stage-wise changes in rice growth, which can be effectively communicated to farmers.

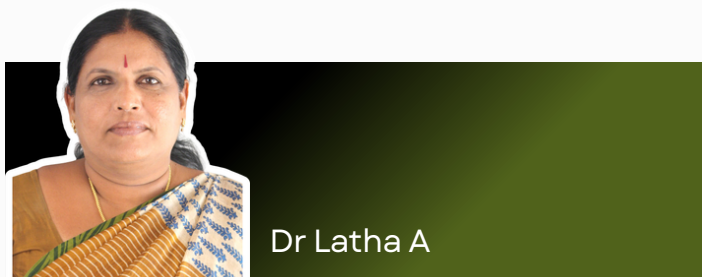
It is anticipated that continued implementation of AWD in the coming years will provide deeper insights into its impact on yield, as well as on pest and disease dynamics in rice cultivation.”

Meet the Team Behind KERA-AWD

Every successful initiative has a story—and at the heart of the KERA-AWD program is a team that brings together science, field experience, and a shared commitment to sustainable agriculture. From decoding data to working closely with farmers and stakeholders, this diverse group ensures that ideas translate into meaningful action on the ground.

In this new section, we take you behind the scenes to introduce the people who make it all happen—the thinkers, planners, problem-solvers and field champions driving KERA-AWD forward. This is just the beginning; while we may not be able to feature everyone at once, we will be gradually introducing all team members through the upcoming issues.

Stay tuned as we continue to spotlight the faces behind work.



Dr. Latha is a Ph.D. holder in Agronomy, specializing in rice agronomy with a focused research background in the unique Kole land ecosystems. She currently serves as Professor and Head at the Agricultural Research Station, Mannuthy, under Kerala Agricultural University (KAU), and also holds the additional charge of Director of Students' Welfare at KAU.

She is the Project Lead for KERA-AWD and serves as the Coordinator of the KERA Cell at KAU. With extensive expertise in sustainable crop production, agricultural mechanization, and enhancing rice productivity under challenging agro-ecological conditions, she has made significant contributions to the field. Dr. Latha has successfully conducted over 30 research projects and has authored more than 75 research publications, reflecting her strong commitment to advancing agricultural science and sustainability.

Dr Prakashan Chellattan
Veettil



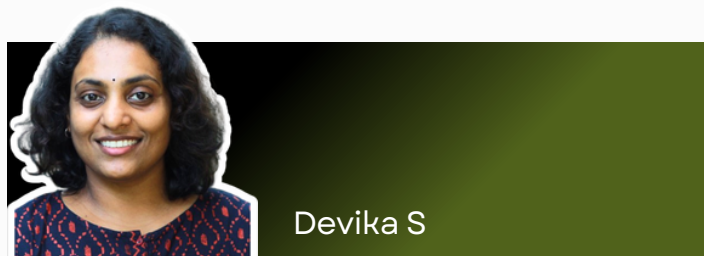
Dr. Veettil is a Senior Scientist at the International Rice Research Institute (IRRI), specializing in agricultural and behavioral economics. With a Ph.D. from Ghent University, he leads major initiatives across South Asia focused on transforming agrifood systems, advancing climate-smart agriculture, and improving farmer livelihoods. His work includes developing innovative crop insurance models, strengthening value chains, and promoting rural entrepreneurship. Dr. Veettil has led key components of CGIAR programs and contributed to policy-oriented research on technology adoption and socioeconomic impact. He has published widely in international journals and is an active member of global agricultural economics associations. He is currently leading the IRRI team in KERA-AWD project.



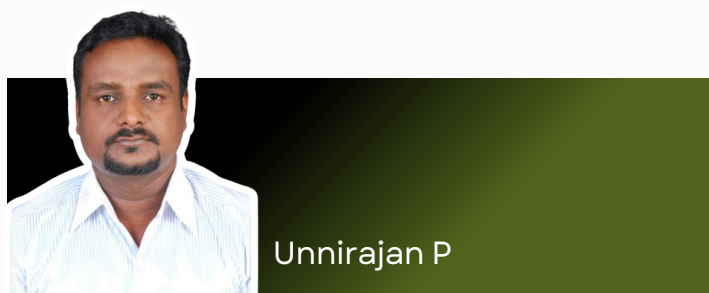
Dr Ashish Chaturvedi

Dr. Chaturvedi is a Senior Scientist and Head of the Training and Outreach Research Group at the Centre for Water Resources Development and Management (CWRDM), Kerala, India. He earned his M.Phil. in Environmental Plant Biology and Ph.D. in Plant Physiology and from the High-Altitude Plant Physiology Research Centre, HNB Garhwal Central University, Uttarakhand, India. With over 13 years of research experience, Dr. Chaturvedi has successfully implemented projects funded by prominent agencies, including the Indian Council of Social Science Research (ICSSR), Department of Science and Technology (DST), Ministry of Jal Shakti (MoWR),

and the Kerala State Council for Science, Technology and Environment (KSCSTE). His current research focuses on climate change adaptation, environmental stress physiology, agricultural water management, and improving crop water-use efficiency in humid environments. Currently, he leads the CWRDM team in the KERA project.



Ms. Devika serves as Lead, Component-1 (Climate Resilient Agriculture) and Agricultural Officer (Procurement) at the State Project Management Unit (SPMU), KERA Project, Government of Kerala, Thiruvananthapuram. In this role, she anchors key initiatives under Component 1, including the implementation of low-emission paddy cultivation in collaboration with leading research institutions and field teams. She holds dual postgraduate degrees in Plant Breeding and Genetics from Cornell University and Tamil Nadu Agricultural University. Her professional journey since 2013 spans roles at Rasi Seeds, VFPC, and the State Department of Agriculture, with experience across research, extension, and farmer engagement.

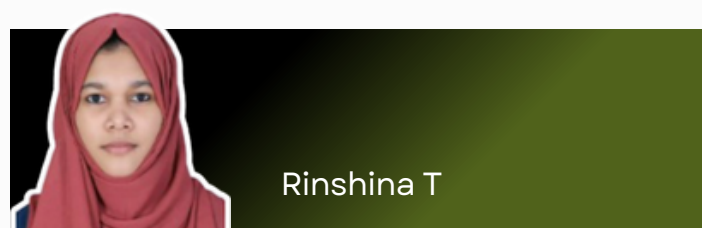


Mr. Unnirajan is the Deputy Director of Agriculture, Department of Agriculture, and is currently working on deputation as the Regional Project Director for the Central Region under the World Bank-assisted KERA project. He coordinates various scheme components of KERA Project across four districts in the Central Region-Ernakulam, Thrissur, Palakkad, and Malappuram.

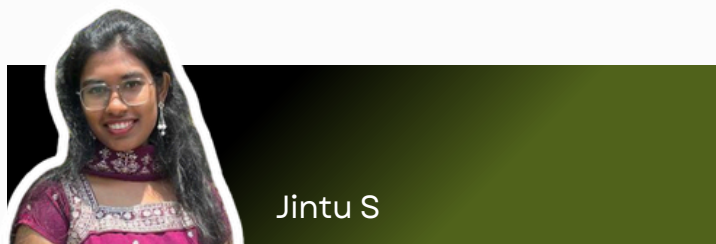


Dr. Divya serves as an Environmental Specialist in the Project Implementation Unit (PIU) of the Kerala Agricultural University (KAU), ARS Mannuthy. In this role, she oversees low-emission paddy cultivation initiatives undertaken collaboratively by IRRI, KAU, and CWRDM, with support from the Department of Irrigation (DoI) and the Department of Agriculture (DoA), and Kerala Agro-Climatic Research Centre (KACRC) activities under Component 1 of KERA. In addition, she holds the charge of Environmental & Social (E&S) Nodal Officer of PIU (KAU).

Dr. Divya earned her Doctorate in Agricultural Economics from Tamil Nadu Agricultural University (TNAU), following her Master's degree in Agricultural Economics and Business Administration (HRM). Her professional journey began in 2004 as an Agricultural Officer in the Department of Agriculture, where she primarily engaged in extension activities supporting farmers. She later transitioned to academia, joining KAU as Assistant Professor in Agricultural Economics.



Ms. Rinshina T is a Project Associate under the KERA AWD project at the Agricultural research Station (ARS), Mannuthy. In her current role, she coordinates extension activities of the project, manages survey data, and undertakes data analysis and related tasks. She completed her B.Sc. in Agriculture from Acharya N.G. Ranga Agricultural University and holds an M.Sc. in Agricultural Economics from Kerala Agricultural University with specialization in irrigation economics. Her academic contributions include publishing research papers in journals and participating in various national conferences and workshops.



Jintu S

Ms. Jintu has completed a Diploma in Agricultural Science from the Institute of Agricultural Technology (IAT), Regional Agricultural Research Station (RARS), Pattambi. Coming from an agricultural family where farming and cultivation form an integral part of daily life, she developed a strong interest in agriculture from a young age. This upbringing, combined with her academic training, has equipped her with both practical experience and theoretical knowledge across various aspects of agriculture and farming practices. She is also one of the skilled field assistants working at the grassroots level, playing a crucial role in supporting data collection and the effective implementation of field activities.

Beyond the Field Notes

The Hut Diaries: IRRI's Kerala Chapter Begins

After much anticipation, IRRI has found a cozy temporary home at the ARS, Mannuthy campus—fondly named “The Hut.” While the permanent office takes shape, this warm little space is already buzzing with ideas, plans, and lively conversations, and those who wish to visit and see the IRRI team can now reach us here. This marks a small yet meaningful beginning in our journey forward.



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